

Airport Design and Simulation Lab



NAME _____

DATE _____

PERIOD _____

Team map • one copy per team

These constraints make a tabletop simulation work. They are not FAA airport-design or operational-training standards.

Team route: ☐ LEGO ☐ Paper ☐ Lucid Team members: _____

JOB	NAME
Planner / route checker	
Mover / materials lead	
Recorder / evidence lead	
Safety checker / timekeeper	

Map checklist: ☐ R1 and R2 do not intersect ☐ Taxi A and B ☐ tower ☐ four gates ☐ north arrow ☐ alternate route

TOP-DOWN SKETCH

Show runways, taxi routes, tower, four gates, north arrow, movement arrows, and one predicted conflict point.

Team readiness and run log · one copy per team

Readiness: Move one token Gate 1 → R1 → a gate. What was blocked or confusing?

Correction before the simulation:

TEST	WHAT HAPPENED?	BREAKDOWN OR EVIDENCE	REVISION BEFORE THE NEXT TEST
Run 1: basic sequence			
Run 2: two requests			
Run 3: R1 closed			

Final breakdown noticed:

Final revision:

Five-step check: ☐ Name ☐ Route ☐ Repeat ☐ Confirm ☐ Log

Roles used: ☐ controller ☐ mover ☐ recorder ☐ safety checker

Speaking, moving, recording, and checking are equal participation routes.

Day 3 individual design note · one copy per student

Word bank: runway · taxi route · gate · conflict point · alternate route

Map feature: _____ Aviation role that uses it: _____

How the feature helps that worker:

Conflict point and the evidence that shows the problem:

Revision completed or still needed:

Use this frame: “A _ *uses the* to _ . *We changed* because the test showed ____.”

Day 3 readiness check

- ☐ My role and work product are both named.
- ☐ My conflict point is visible on the team map.
- ☐ My revision explains what would change.

Day 4 individual evidence

One copy per student

Use this frame: “Our goal was _ *during the* -minute block. The log shows _, *so next we would*.”

Specific improvement goal:

Exact two- or three-minute block used: _____

Person, tool, or evidence that supported the change:

Evidence the change helped or did not help:

Next adjustment if the scenario ran again:

New scenario

Alpha and Bravo request movement at the same time. Taxi A is blocked.

Aircraft moving first and reason:

Full five-step classroom call:

One two-minute team goal before the next run: _____